Presentation 5

- Links to presentation(s) and code(s) on GitHub
    - 12/8 Presentation Link
    - Menu Traffic Dashboard link: (1) web, (2) pbix on Github
    - Code: CAR data (unchanged since previous presentation)
        - Data for Main Menu Traffic Dashboard: jupyter notebook
            - Based on Krish's data transformations for the sub-menu traffic, I transformed the data starting at the main menu
        - Data for the Sub-Menu Traffic Dashboard: Krish's jupyter notebook
            - I ran this code to generate the data needed to recreate the time trends component of Krish's dashboard, which focused on some key menus following the legal menu.
- What did you do?

  First, I combined Katherine and I's dashboards into one dashboard. After this, I analyzed my portion (main menu trends) for relevant key insights, building on previous week's analysis.

How does this help the project?

With a single dashboard, Legal Aid users can efficiently navigate between visuals. Additionally, I made my recommendations more streamlined so that they can be more realistically applied.

- Issues faced (if any)
- Attempts to resolve issues (if any)
- Next Steps

Presentation 4

- Links to presentation(s) and code(s) on GitHub
    - 11/19 Presentation Link
    - Menu Traffic Dashboard links: (1) web, (2) pbix on Github
    - Code: CAR data (unchanged since previous presentation)
        - Data for Main Menu Traffic Dashboard: jupyter notebook
            - Based on Krish's data transformations for the sub-menu traffic, I transformed the data starting at the main menu
        - Data for the Sub-Menu Traffic Dashboard: Krish's jupyter notebook
            - I ran this code to generate the data needed to recreate the time trends component of Krish's dashboard, which focused on some key menus following the legal menu.
- What did you do?

  First, I combined Katherine and I's dashboards into one report. After this, I created a seasonal trend tab for the legal menu, as I noticed that this was missing. I then created buttons to toggle between all of these pages so that the user can easily move between tabs. Then, I added a date range slicer so that the user could choose the date range they wanted to filter calculations for. I also cleaned up the graphics to only display the Top N number of most popular sub-menus. I then analyzed my dashboards for any notable and relevant trends.

- How does this help the project?

  This provides Legal Aid with a much more concise and efficient way to maneuver between visuals and enhances the readability and clarity of visuals. Additionally, I uncovered some interesting and relevant trends from the seasonal legal trends dashboard that I created this week.

- Issues faced (if any)

  I attempted to organize the legend by which lines appeared highest visually but couldn't find an effective way to do this dynamically. The only solutions I was able to conjure would order them statistically, which wouldn't work when we have different filters when drilling down.

- Attempts to resolve issues (if any)
- Next Steps

The large increase in family menu calls is of particular interest to me, particularly those ending in a closed queue. I would like to uncover what types of issues specifically are driving this increase in call volume, and may look into the All Calls data to find this information at a more granular level.

Presentation 3

- Links to presentation(s) and code(s) on GitHub
    - 11/6 Presentation Link
    - Dashboard Links:
        1. Main Menu Traffic Dashboard (web)
            - Pbix file on Github found here.
        2. Sub-Menu Traffic Dashboard (web)
    - Code: CAR data
        1. Data for Main Menu Traffic Dashboard: jupyter notebook
            - Based on Krish's data transformations for the sub-menu traffic, I transformed the data starting at the main menu
        2. Data for the Sub-Menu Traffic Dashboard: Krish's jupyter notebook
            - I ran this code to generate the data needed to recreate the time trends component of Krish's dashboard, which focused on some key menus following the legal menu.

- What did you do?

  First, I used Krish's code for the legal sub-menus (Family, Housing, etc) and replicated the hourly time trends drill-down dashboard (presented by Katherine). Then, Based on suggested next steps from Krish, I transformed the data In a similar way, starting from the main menu, to provide a more "zoomed out" picture of call flow trends over time. I conducted analysis at varying degrees of granularity at the hourly, weekly, and monthly level. This is reflected in the Main Menu Traffic Dashboard linked above.

- How does it help the project?

  I was able to identify how trends varied for seniors compared to non-seniors, as well as how the legal issues that they called about had differing patterns over time. If Legal Aid has goals about the proportion of seniors vs. non-seniors, as well as the legal issue types that they aim to serve in a given month or year, this can inform how they should allocate resources and staff to serve their target clientele.

- Issues faced (if any)
- Attempts to resolve issues (if any)
- Next Steps

  For my next steps, depending on instructor and Cynthia's feedback, I would be interested in exploring how queue and callback success rates vary for seniors vs. non-seniors over time, specifically over the course of the day. I have a hypothesis that since non-seniors have less flexibility and more additional commitments and constraints, they may be less likely to reach an intake specialist depending on what time they are called back. I can do this by looking at the activity name column in the CAR data to parse callbacks and whether or not they were successful.

- References (Mention if you built up on someone else's work)

  I used Krish's data transformation code that he use for his presentation with Cynthia last week. This notebook also inspired the data transformations that I did to get the data used in my main menu call flow dashboard.

Presentation 2 (UPDATED)

- Links to presentation(s) and code(s) on GitHub
  - Presentation link (original from 10/21)
  - Dashboard link (updated based on 10/24 meeting)
  - Code: CAR data
    1. Initial data import used jupyter notebook code provided by Krish.
    2. Initial data cleaning conducted in this R script
       - New dataset of queue wait times constructed from CAR data included in the updated R script.
    3. Transformed dataset with menus that queue originated from in this R script.

- What did you do?

I first transformed the CAR call activity data into a clean session-level dataset in order to classify caller outcomes and intake pathways. First, I identified which callers actually reached an endpoint and which reached an open intake queue, so I could distinguish real intake opportunities from incomplete calls. I then filtered out LAC callback traffic, which does not follow the standard intake flow. For remaining sessions, I extracted the most recent menu a caller reached and standardized it into a unified **PrimaryMenu** category (collapsing variations of senior-related menus into “Senior Menu” and some legal paths into “Legal Menu”). I kept the session’s queue and earliest timestamp. This produces a concise and analyzable structure showing who reached intake, through which menu, and from which queue, enabling clear downstream intake analytics.

- How does it help the project?

This enabled me to use drill-through features in PowerBi so that after identifying menu-level trends in calls, we can identify queue-level trends within menus.

- Issues faced (if any)

- Attempts to resolve issues (if any)

- Next Steps

Based on Krish/Cynthia's feedback, I can analyze time trends for different menus and queues at the hourly level. I particularly would like to see if Seniors have a higher proportion of calls that make it into the queue vs. calls that end in closed queues as a result of having more flexibility. This will require some modifications of the dataset so that I create a flag for calls that ended in a closed queue.

- References (Mention if you built up on someone else's work)

Presentation 2

- Links to presentation(s) and code(s) on GitHub
    - Presentation link
    - Dashboard link
    - Code: CAR data

1. Initial data import used jupyter notebook code provided by Krish.
2. Initial data cleaning conducted in this R script
    - New dataset of queue wait times constructed from CAR data included in the updated R script.

- What did you do?

For Presentation 2, building off of last week's queue analysis, I looked at when callers were reaching the open queue vs. getting a close queue message when they called during LAC's business hours. Additionally, I explored the distribution of wait times in the queue for callers who were able to enter the open queue and got a call back.

- How does it help the project?

Cynthia mentioned how she was wondering if it made more sense to open the queue again during the middle of the day. This project helps to inform her of when those who are hoping to reach a queue tend to call, and when the queues tend to close.

- Issues faced (if any)

I initially thought that the wait time would be telling of how long someone was waiting even if they gave a callback number and hung up, but base on my exploration of the data, it seems that these people do not have the same contact session ID when they are called back, thus it is difficult to understand how long they are waiting for a call back for.

- Attempts to resolve issues (if any)

I am going to ask Krish/Cynthia for clarifications on this to inform my next steps.

- Next Steps

If the situation with the hang up/callback being a different contact session ID is true, I will try to analyze the success rates of these callbacks. If they have a low success rate, this may indicate that callers feel discouraged by long queue wait times. Furthermore, I want to analyze the paths that people who reach an open queue take vs. those who end up reaching a closed queue. I am wondering if seniors who are more likely to not have to work, are the ones who are able to reach an open queue whereas those who have a strict work M-F schedule/other responsibilities are unable to catch the open queue. I plan on doing this using the CAR dataset and tracing the flow of the call before they reach either a closed queue or open queue.

- References (Mention if you built up on someone else's work)

Presentation 1

- Links to presentation(s) and code(s) on GitHub
  - Presentation link
  - Dashboard link
  - Code: CAR data transformations
    1. Initial data import used jupyter notebook code provided by Krish.
    2. All data transformations conducted in this R script.
- What did you do?

I visualized how queue volume evolved over time by month using the CAR data, analyzing how volume by legal issue type was driving these variations in volume over time. I excluded the subsenior queues and transfers, since I wanted to understand the volume and demography of clients who needed intake specialists.

- How does it help the project?

I thought it was important to develop a baseline understanding who is calling with what types of issues in the first place seeking help and how their needs are evolving in a dynamic legal and political landscape.

I chose to analyze the queues because I feel like after the phone system "triaged" callers, the people who made it all the way to be placed on a queue represent those who feel that their issue can be helped by LAC in some capacity and want to speak to a live intake specialist. This filters the best "match" of potential clients for LAC/where LAC can maximize their impact.

- Issues faced (if any)

Queues do not represent an "endpoint" for all calls, as many callers do not ultimately get placed into a queue either because they end the call before then either due to frustration or their issue isn't serviced by LAC, or the queue is closed due to capacity constraints, or they call outside of clinic hours. Thus, they do not fully capture the distribution of all caller needs.

- Attempts to resolve issues (if any)

In my next steps, I will conduct more granular analysis and try to glean insights about callers who do not ultimately end up being placed in a queue due to the aforementioned reasons.

- Next Steps

Initially in my presentation, I highlighted 3 potential next steps: (1) analysis of closed queue volume throughout the day, queue wait time analysis, and callback analysis. Following presentation feedback, I will focus my efforts next week first on (1) analysis of closed queue volume and later focus on (2) queue wait time analysis, as I feel like these go hand-in-hand when determining whether or not to close the queue earlier, and then plan on reopening it later in the day, potentially around midday or the early afternoon.

The EP column in the CAR data should make analyzing the volume of calls that end up in closed queues fairly transparent when we filter for LAC's business hours/days. This analysis can be conducted at the hourly or even 30-minute interval level, to look at both volume and proportion of calls throughout the day that end up being routed to the closed queue.

- References (Mention if you built up on someone else's work)

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